5 A boy looks at the image of a clock in a plane mirror. Fig. 5.1 shows the mirror, the clock and the position of one of the boy's eyes.

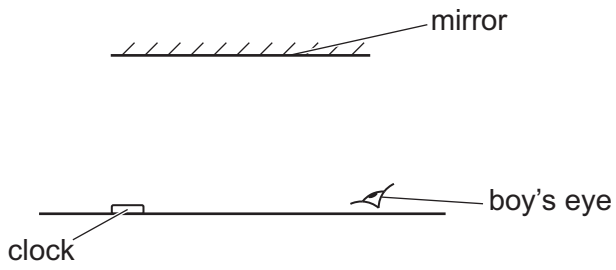


Fig. 5.1

(a) (i) On Fig. 5.1, draw a ray of light from the clock, reflected to the boy's eye. [2]

(ii) On Fig. 5.1, mark with an X the position of the image of the clock. [1]

(iii) State whether the image formed by the mirror is virtual or real.

Explain your answer.

.....

..... [1]

(iv) Fig. 5.2 shows the image of the clock seen by the boy.

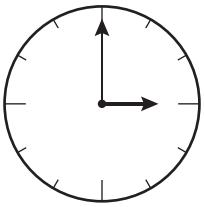


Fig. 5.2

The boy now looks directly at the clock.

On Fig. 5.3, draw what the boy sees.

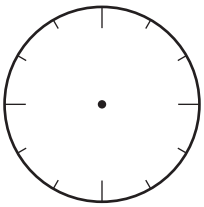


Fig. 5.3

[1]

(b) (i) The clock is illuminated by a source of monochromatic green light.

State the meaning of monochromatic.

..... [1]

(ii) The green light has a wavelength of 5.6×10^{-7} m.

Calculate the frequency of this green light.

frequency = ... [3]

[Total: 9]

© UCLES 2022